Lab 2 – Lakehouse Federation & Foreign Catalog in Databricks

Lab Summary

In this lab, we learn how to create a **Lakehouse Federation connection** in Databricks.

We will:

- Create an external connection
- Create a federated catalog
- Understand foreign catalogs
- Understand foreign tables
- Understand when to use Lakehouse Federation

This lab is important for querying external databases (like MySQL, Redshift, SQL Server, etc.) without migrating data into Databricks.

Prerequisites

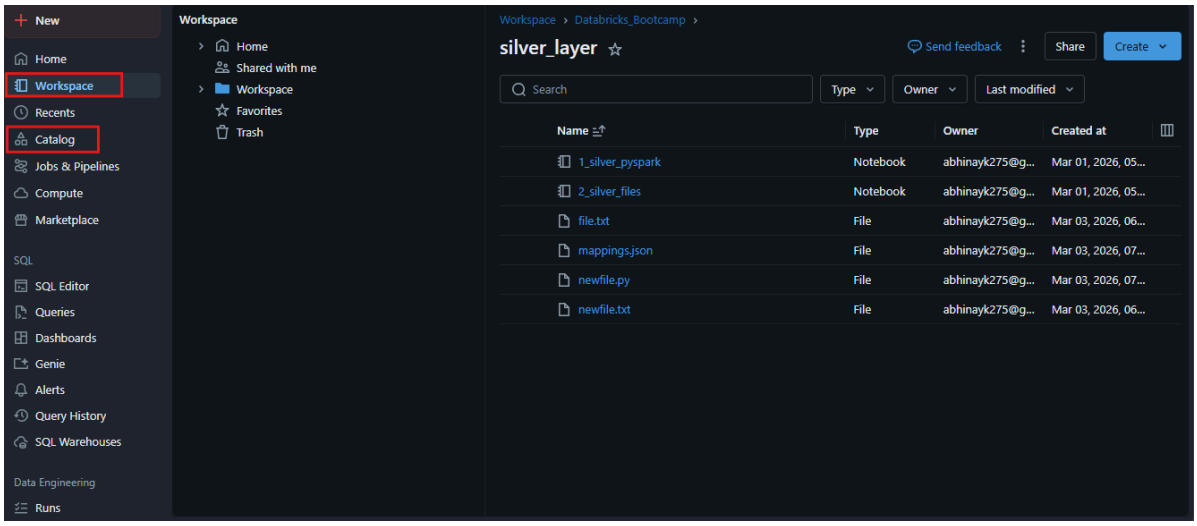
- Databricks workspace access
- Unity Catalog enabled
- Required permissions to create connections
- External database (MySQL / Redshift / SQL Server) – optional for demo

Steps to Complete Lab

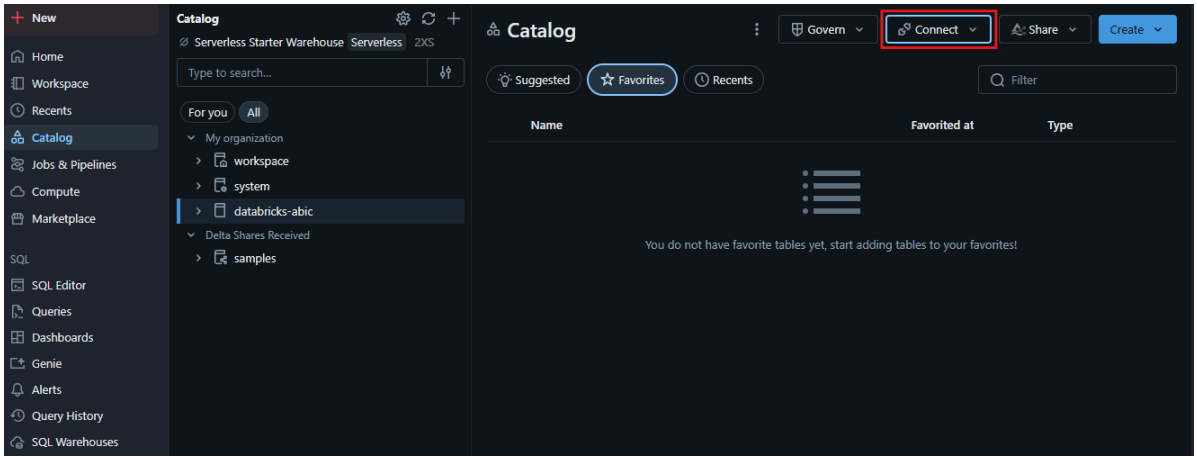
Task 1 – Navigate to External Data Section

1.1 Open Catalog

1. Go to **Databricks Workspace** —→ Click on **Catalog**

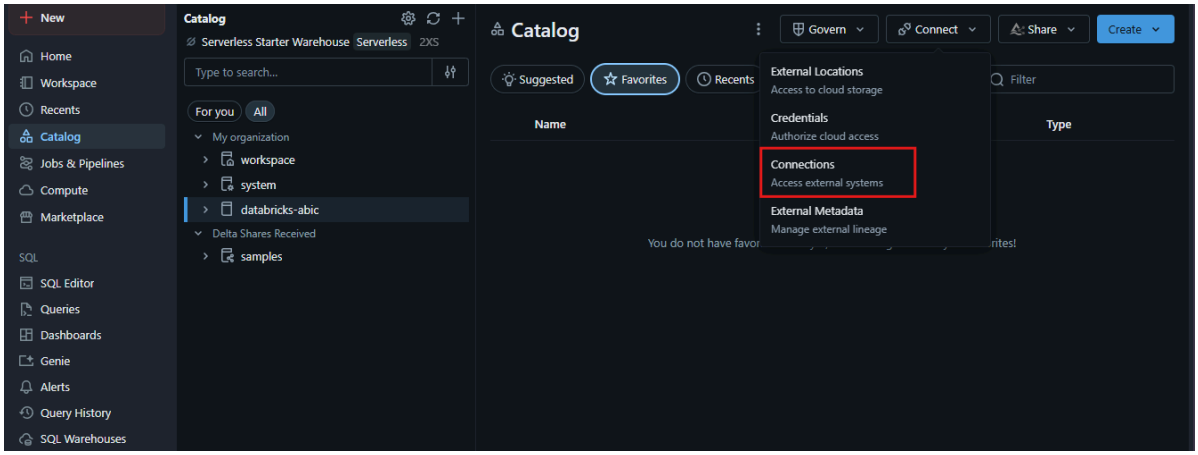


2. Click on **Connect**

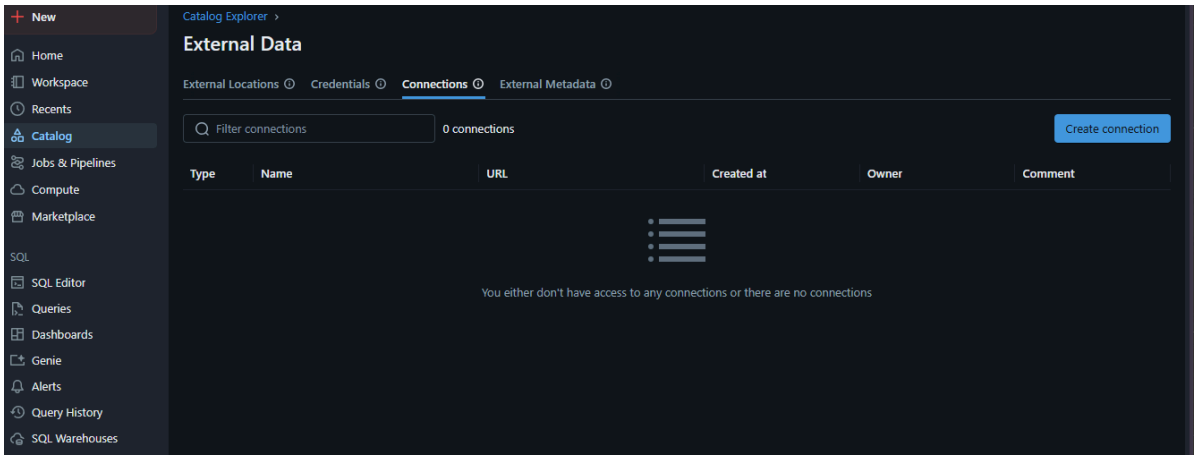


1.2 Open Connections

1. Click on **Connections**



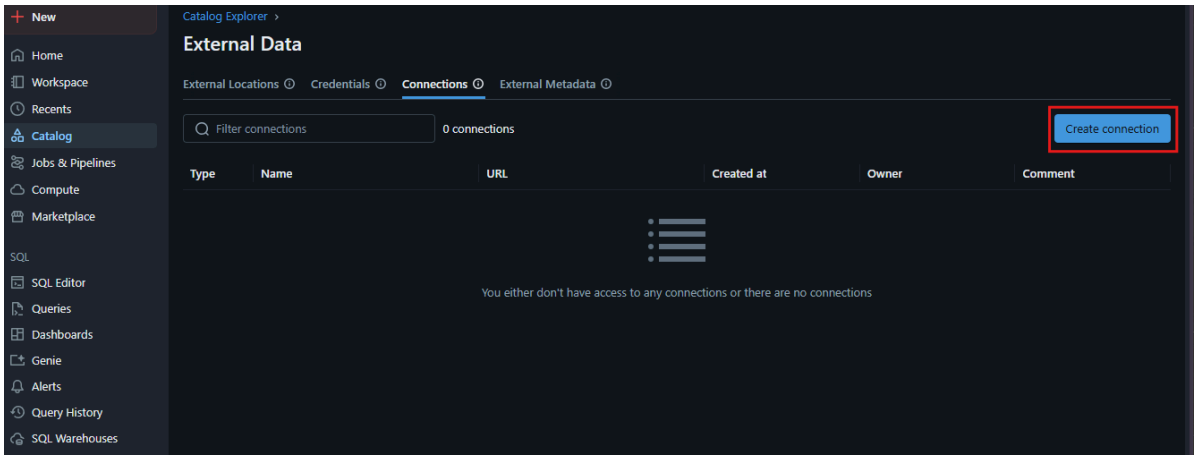
2. You will see the list of existing connections
3. If none exist → list will be empty



Task 2 – Create a New Connection

2.1 Click Create Connection

1. Click **Create Connection**

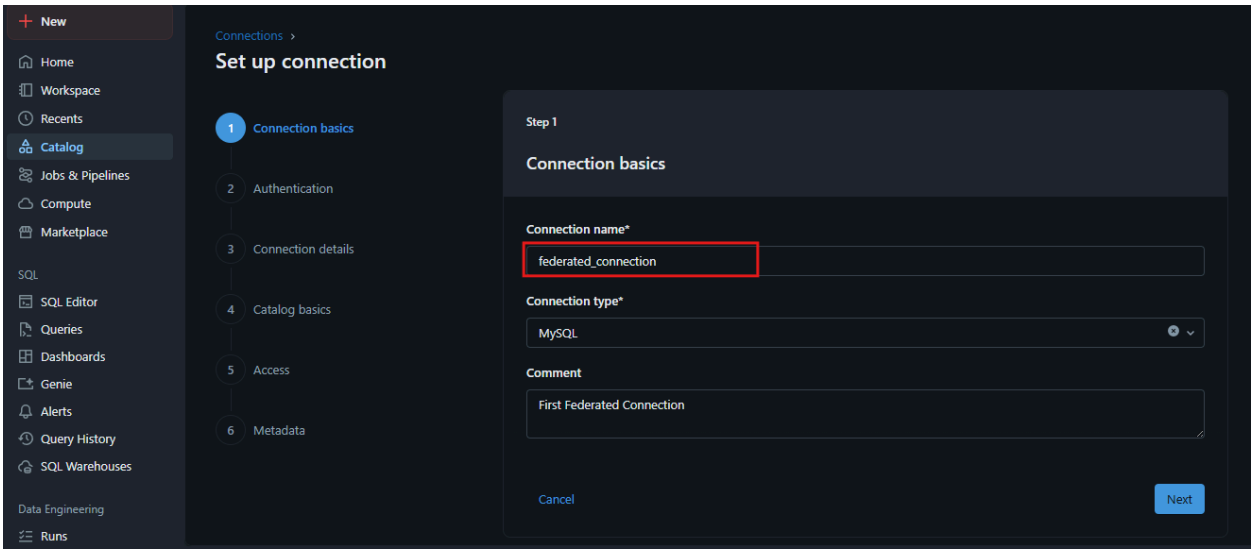


2.2 Provide Connection Details

Connection Name

Example:

```
federated_connection
```



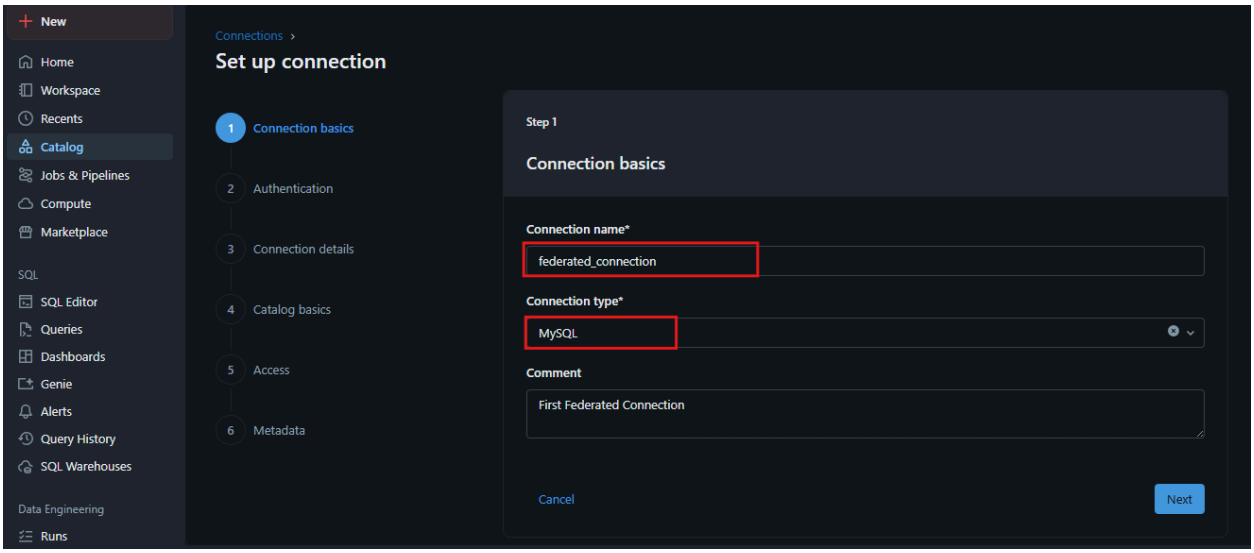
⚠️ Important:

This name will also be used as the catalog name.

Connection Type

Select database type:

For demo → select **MySQL**

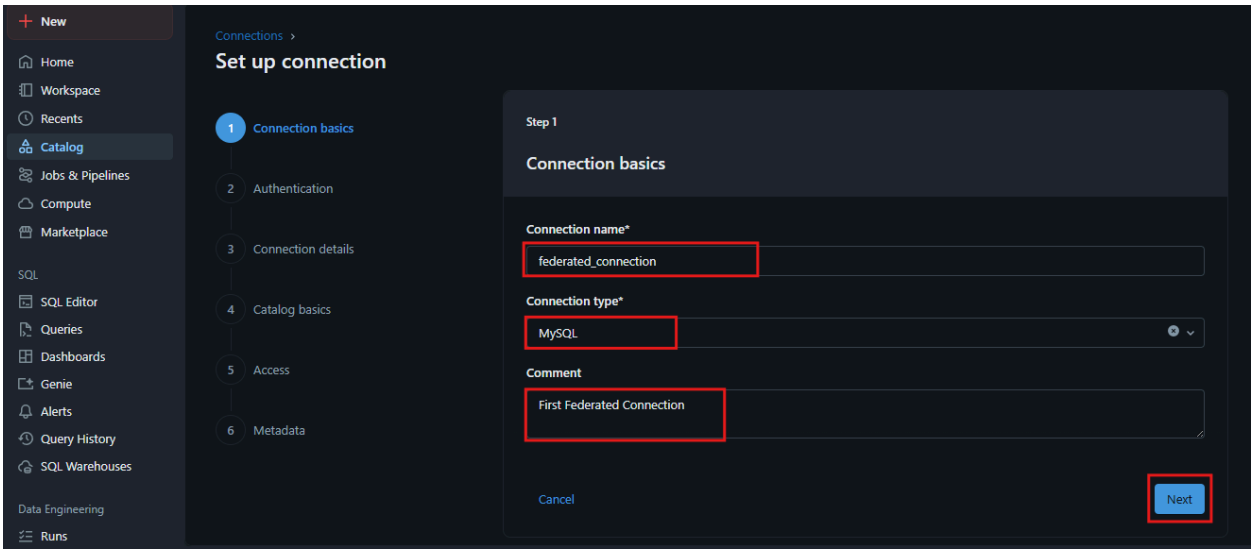


Add Comment (Optional)

Example:

```
First Federated Connection
```

Click **Next**



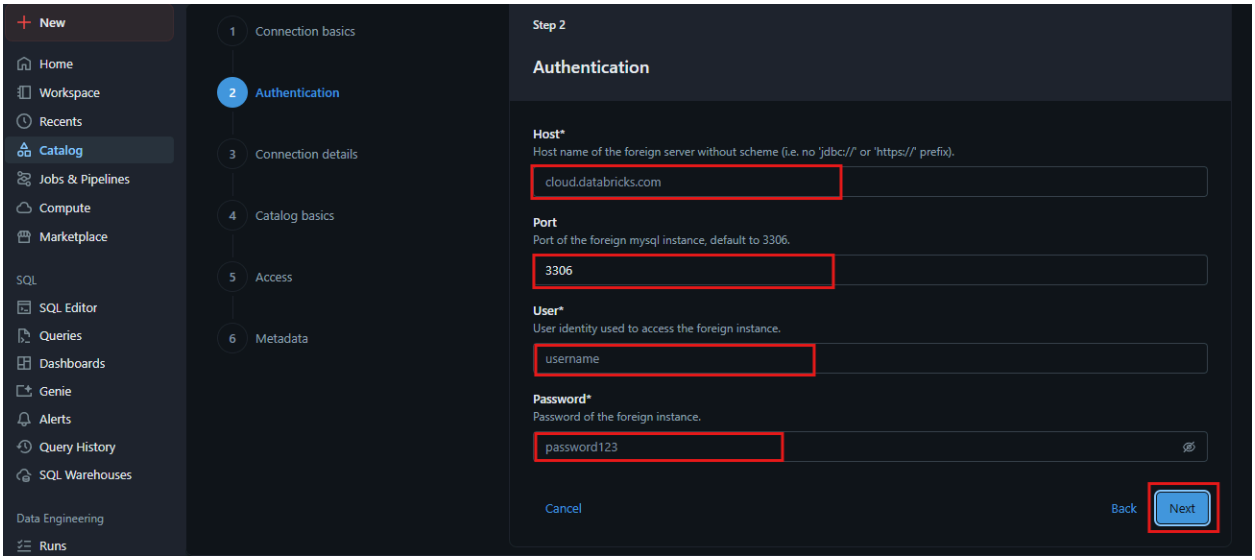
2.3 Provide Database Credentials

Provide:

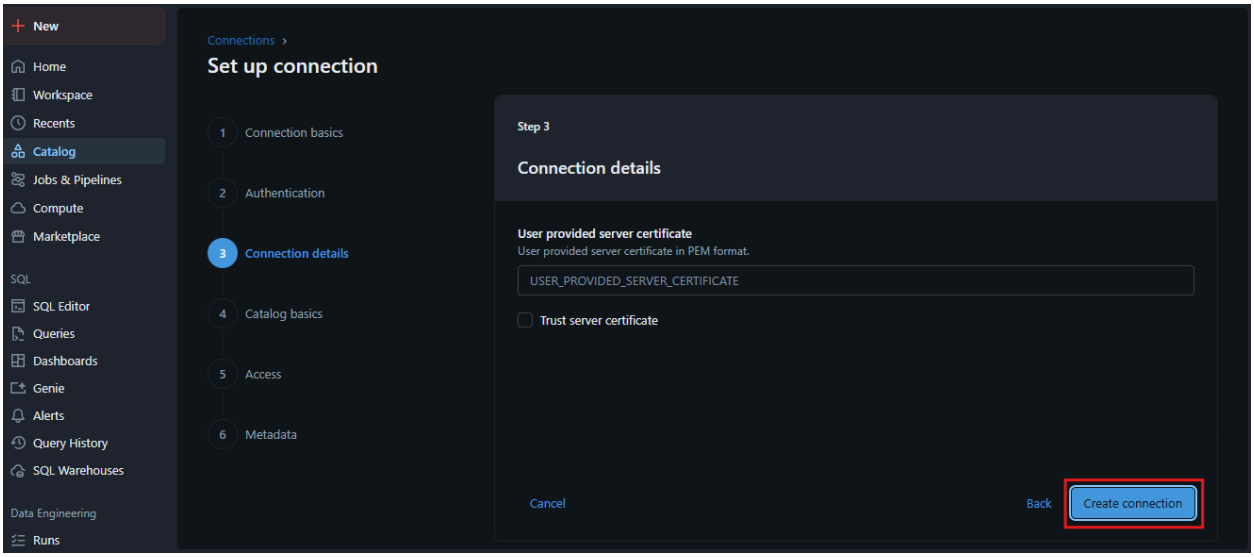
- Hostname :
- Port : 3306
- Username :
- Password :

For demo purpose you can use dummy values.

In real-world → database must be hosted and accessible.



Click Next —→ Click **Create Connection**

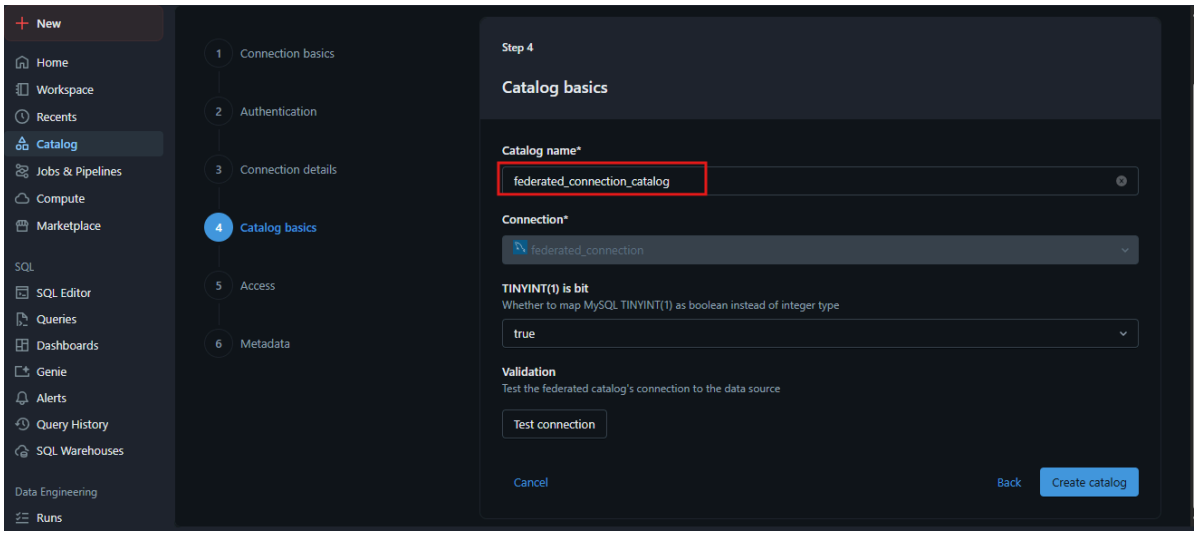


Task 3 – Create Federated Catalog

After creating connection:

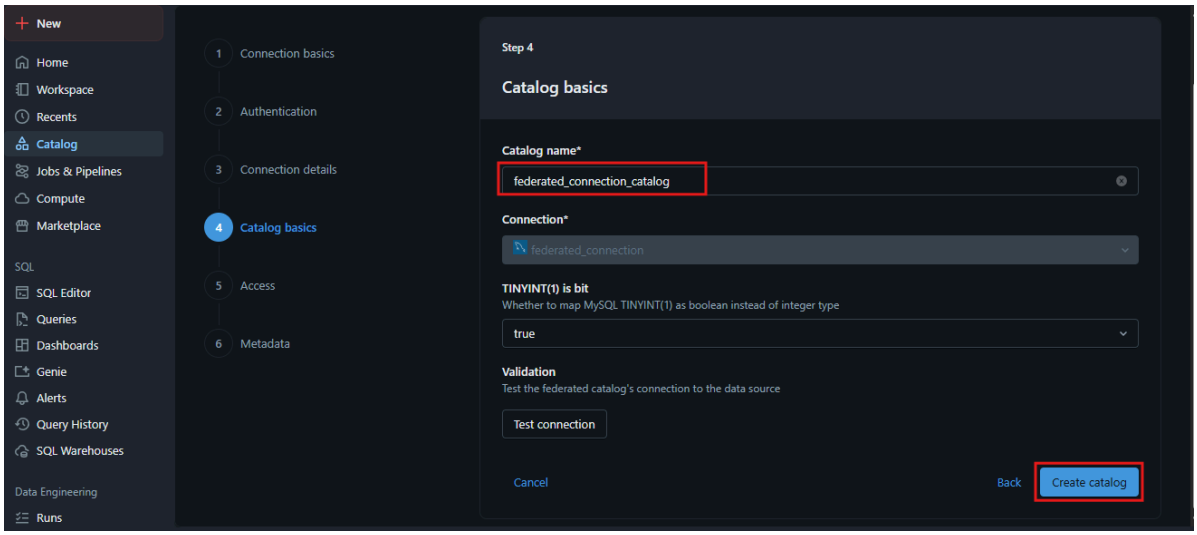
1. Databricks will prompt to create a catalog
2. By default:

```
federated_connection_catalog
```



3. You can rename if required

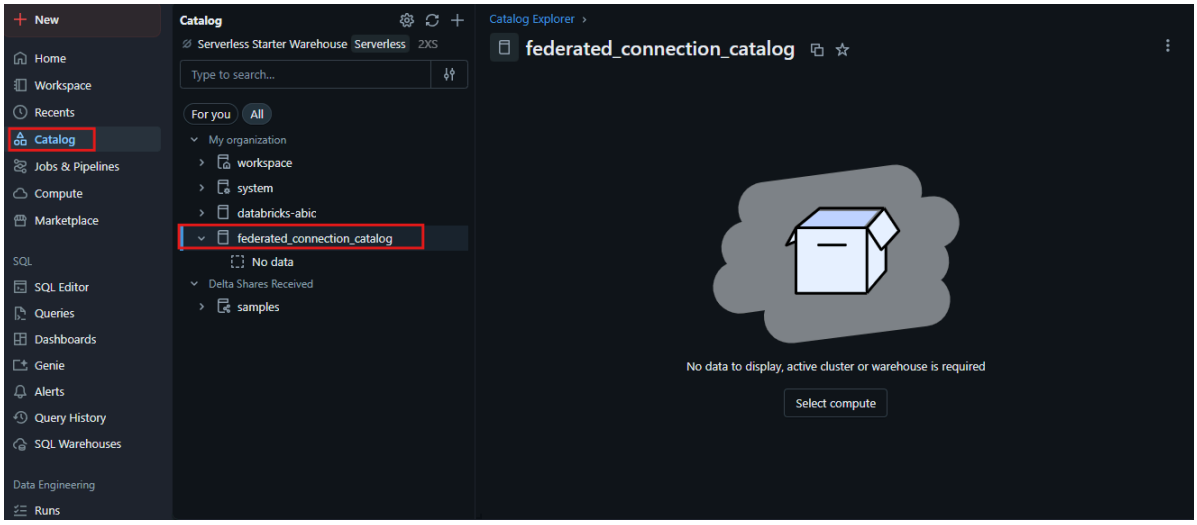
4. Click **Create Catalog**



5. Click **Next —→ Save**

Task 4 – Verify Catalog Creation

1. Go to **Catalog**
2. You will see your new catalog



Outcome

After completing this lab, you can:

- Create federated connections
- Create foreign catalogs
- Understand foreign tables
- Use Lakehouse Federation for POC
- Avoid unnecessary data migration